

Exp No. 4: OFDMA Resource Block Allocation

AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access) resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

Objective 1: To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.

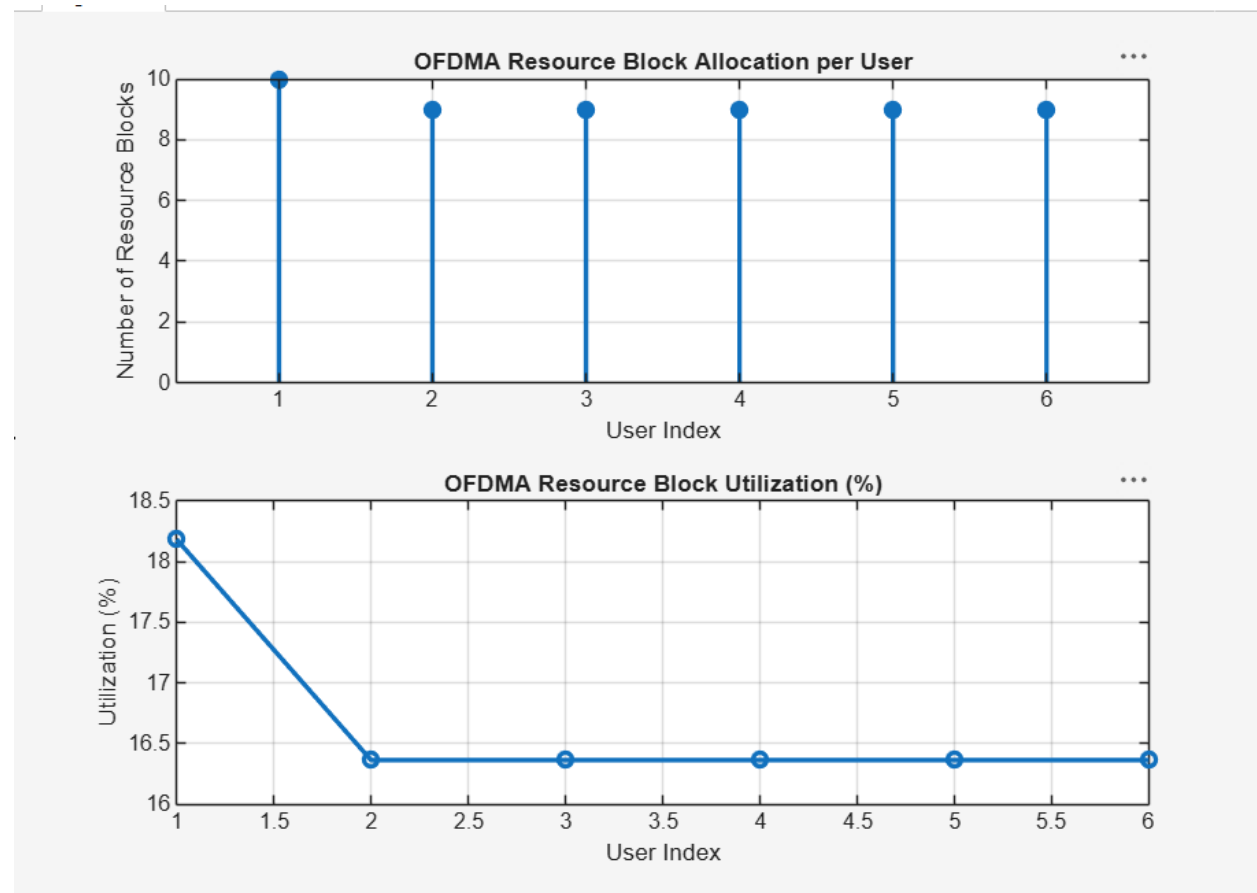
Program

```
clc;
clear;
close all;
sysBW = 10e6;
rbWidth = 180e3;
resourceBlocks = floor(sysBW/rbWidth);
userID = 1:6;
rbPerUser = floor(resourceBlocks/length(userID));
rbAllocation = repmat(rbPerUser,1,length(userID));
extraRB = mod(resourceBlocks,length(userID));
rbAllocation(1:extraRB) = rbAllocation(1:extraRB) + 1;
rbUsage = (rbAllocation/resourceBlocks) * 100;
disp('Total Resource Blocks:')
disp(resourceBlocks)
disp('RB Allocation per User:')
disp(rbAllocation)
figure
subplot(2,1,1)
stem(userID,rbAllocation,'filled','LineWidth',2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
```

```

grid on
subplot(2,1,2)
plot(userID,rbUsage,'-o','LineWidth',2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

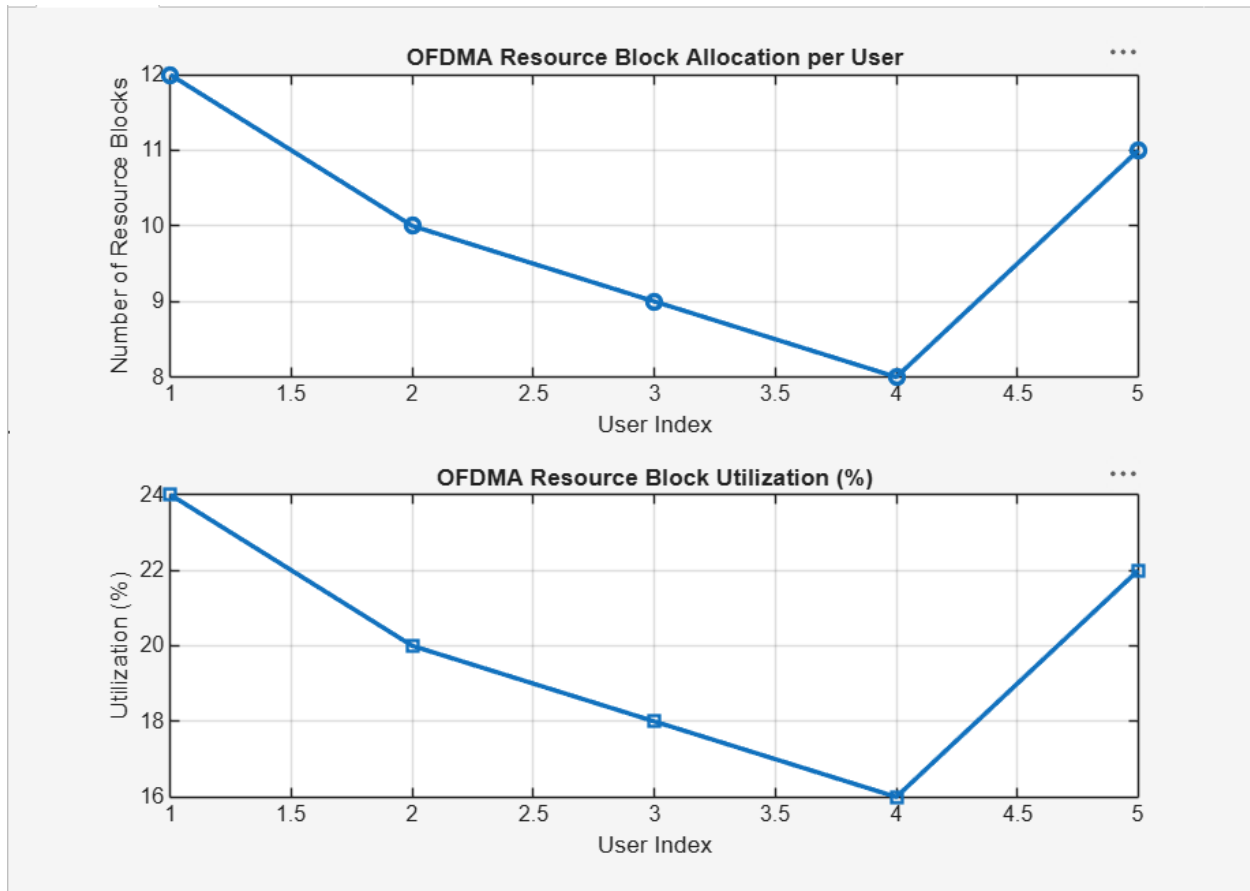
```



Objective 2: To evaluate the **Resource Block Utilization (%)** by analyzing how efficiently available RBs are assigned to active users.

Program

```
clc;
clear;
close all;
rbTotal = 50;
userIndex = 1:5;
rbAssigned = [12 10 9 8 11];
usagePercent = (rbAssigned ./ rbTotal) * 100;
disp('Resource Block Utilization (%) per User:')
disp(usagePercent)
figure
subplot(2,1,1)
plot(userIndex,rbAssigned,'-o','LineWidth',2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
subplot(2,1,2)
plot(userIndex,usagePercent,'-s','LineWidth',2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

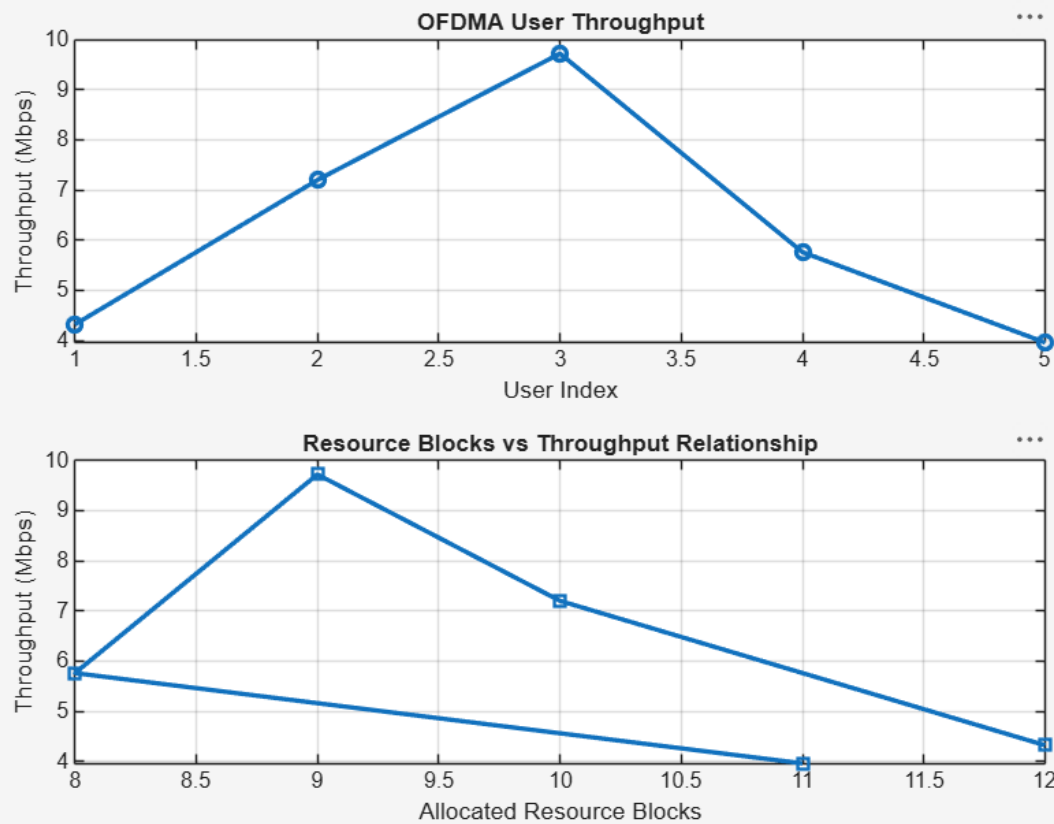


Objective 3: To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

program

```
clc;
clear;
close all;
rbSize = 180e3;
userID = 1:5;
rbCount = [12 10 9 8 11];
spectralEff = [2 4 6 4 2];
userRate = rbCount .* spectralEff .* rbSize;
userRate = userRate / 1e6;
disp('User Throughput (Mbps):')
```

```
disp(userRate')
figure
subplot(2,1,1)
plot(userID,userRate,'-o','LineWidth',2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
subplot(2,1,2)
plot(rbCount,userRate,'-s','LineWidth',2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```



Result:

- 1.The propagation loss was successfully computed for Sub-1 GHz, Mid-band, and mmWave frequency bands using the Free Space Path Loss (FSPL) model.
- 2.It was observed that propagation loss increases with increase in operating frequency, with mmWave showing the highest attenuation and Sub-1 GHz showing the lowest loss.
- 3.The comparative analysis of spectrum bands clearly demonstrates that lower frequency bands are more suitable for long-range communication, while higher frequency bands require advanced techniques like beamforming for efficient performance.